

VALOGNES, France

WMO: 070390

Lat: 49.517N

Lon: 1.500W

Elev: 200

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.0	30.0	23.2	17.6	32.0	25.8	19.9	33.5	32.0	48.5	27.7	47.5	2.9	10	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	78.0	67.1	74.6	65.4	71.6	63.3	68.1	76.3	66.1	72.7	64.5	70.0	6.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	93.0	71.9	63.4	87.9	69.4	62.0	83.7	67.2	32.4	76.4	30.9	73.0	29.6	70.4	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	18.3	15.7	DB	22.6	84.4	2.9	5.0	20.6	87.9	18.9	90.9	17.2	93.7	15.1	97.3	(4)
(5)			WB	22.1	71.9	2.8	2.9	20.1	73.9	18.4	75.6	16.8	77.2	14.8	79.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	43.0	43.0	45.1	49.1	54.5	59.1	61.8	62.3	59.0	53.6	47.6	43.2	
	DBStd	8.70	5.55	5.31	5.29	4.94	4.58	4.22	3.91	3.85	4.54	5.13	4.87	5.80	
	HDD50	1022	221	201	165	74	11	0	0	0	0	25	103	222	
	HDD65	4858	681	616	616	476	327	184	116	102	187	353	523	677	
	CDD50	1687	5	5	14	47	149	273	366	380	271	137	30	10	
	CDD65	49	0	0	0	0	1	7	17	17	7	0	0	0	
	CDH74	338	0	0	0	0	14	56	116	111	40	1	0	0	
	CDH80	59	0	0	0	0	1	8	15	31	4	0	0	0	
Wind	WSAvg	7.2	8.5	8.7	8.8	8.2	7.6	6.5	6.4	5.8	5.5	6.0	6.5	8.1	
Precipitation	PrecAvg	35.5	4.0	3.0	2.4	2.4	2.0	1.9	1.9	2.3	2.4	4.0	4.5	4.7	
	PrecMax	45.6	7.2	6.4	6.2	6.1	4.5	4.4	4.1	5.9	4.8	8.5	11.3	10.0	
	PrecMin	25.2	0.7	0.5	0.5	0.3	0.4	0.5	0.6	0.4	0.4	1.1	1.3	0.6	
	PrecStd	5.1	1.9	1.7	1.3	1.5	1.1	1.1	0.9	1.4	1.1	1.9	2.2	2.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	56.7	61.5	68.7	75.8	79.6	82.6	84.3	79.1	70.1	60.9	56.6	
		MCWB	52.8	51.1	53.4	58.1	64.7	67.2	67.3	73.9	66.5	62.4	58.4	54.2	
	2%	DB	53.9	53.9	58.1	63.7	70.1	74.7	77.9	76.3	73.6	65.0	58.2	54.8	
		MCWB	51.8	50.7	52.3	55.1	61.0	65.3	66.6	67.3	65.0	60.1	55.2	53.0	
	5%	DB	52.2	52.6	55.3	60.1	66.4	70.8	73.9	72.9	70.1	62.8	56.6	53.5	
		MCWB	50.2	49.9	51.3	53.5	59.2	62.3	64.4	65.0	63.1	59.0	53.8	51.6	
	10%	DB	50.9	51.2	53.2	57.5	62.8	68.1	70.7	70.0	66.9	61.2	55.2	52.2	
		MCWB	49.0	48.8	50.1	52.1	56.9	60.7	62.5	62.8	61.4	58.1	52.7	50.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	53.0	55.2	59.4	65.9	69.0	69.8	74.0	68.3	63.8	59.6	55.1
			MCDB	54.6	55.0	59.7	65.6	74.8	77.2	79.5	83.6	75.8	67.5	60.8	55.8
2%		WB	52.2	51.7	53.3	56.3	61.9	65.8	67.3	68.1	65.7	61.2	56.1	53.5	
		MCDB	53.4	53.2	56.2	62.1	68.6	73.4	76.1	75.4	71.4	63.7	57.5	54.5	
5%		WB	50.7	50.5	52.0	54.2	59.6	63.3	65.3	65.7	64.0	59.9	54.5	52.0	
		MCDB	51.9	52.1	54.5	58.8	65.3	69.2	72.4	71.2	68.5	62.1	56.0	53.4	
10%		WB	49.3	49.0	50.5	52.7	57.6	61.6	63.5	64.0	62.5	58.6	53.0	50.1	
		MCDB	50.7	50.8	52.8	56.7	62.0	66.7	69.1	68.8	65.8	60.7	54.8	51.9	
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.0	11.5	13.3	13.6	15.1	14.1	14.5	15.1	12.5	11.3	9.4	
		MCDBR	8.0	10.6	14.6	18.1	19.8	20.3	21.2	19.7	19.6	14.1	11.3	8.3	
		MCWBR	7.5	8.4	10.1	11.0	12.4	12.2	11.8	11.3	11.4	10.2	9.1	8.4	
	5% WB	MCDBR	7.2	8.8	12.5	16.2	18.3	18.3	18.6	18.2	17.1	12.4	10.3	8.1	
		MCWBR	7.3	8.1	9.4	10.3	12.0	11.8	10.8	11.2	10.5	9.6	9.2	8.4	
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.381	0.393	0.399	0.395	0.389	0.383	0.372	0.361	0.336	0.326		
	taud	2.412	2.395	2.309	2.296	2.329	2.371	2.388	2.426	2.438	2.454	2.443	2.380		
	Ebn at Noon	226	251	260	270	273	275	274	271	262	243	222	208		
	Edh at Noon	21	26	34	38	38	37	36	33	30	25	20	19		
All-Sky Solar Radiation	RadAvg	280	510	891	1408	1664	1852	1787	1479	1132	636	341	227		
	RadStd	25	61	95	119	126	179	114	99	70	52	42	28		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page